

and already proven in the processor selected.

What needs to be factored

When choosing the architecture (both hardware and software), designers need to be conscious of three critical aspects:

1. Cost of the product
2. Volume of the product to be produced
3. Types of users and the working environment of the product.

Let us briefly see each of them for better understanding.

Cost. It is the primary criteria for the success of a product. Typically, the cost of a product equals Bill of Material + Manufacturing Charges + Warranty & Product Support cost. Please note, this cost is not the final price, which would include other expenses like packaging and marketing and profit margin. When an off-the-shelf design is used, it is this cost that gets impacted. Typically, 70% will be BoM, 20% will be manufacturing, and 10% will be other cost. From this you can see how a proper design can save the cost as well as ensure the success of the product.

Sales volume of the product.

With higher product volumes, the manufacturing process has to be automated, and the design has to be robust enough for fast and reliable manufacturing. Most contract manufacturers use processes like Design for X (where X stands for assembly, manufacturing, testing, compliance, and environment). An off-the-shelf design makes these processes difficult as reference design based products make the DFT and DFC difficult.

User and working environment.

Third important aspect is the users who use the product and the product's working environment. About 50 to 60% features of a product depend on these two parameters. When a predesigned (reference) circuit is used, customisation for the user to make the product work becomes difficult in environment-prescribed scenario. A harsh environment needs special parts (industrial or military grade) and a robust thermal design mechanism, not to mention radiated emission (like EMC/EMI) issues that

are dependent on the components and the circuit design.

It does not mean you should not use off-the-shelf designs at all; there will be situations where it may become unavoidable. In case of PC motherboards, for example, the designs are typically provided by the chip vendors like Intel. If a customer wants to customise the design, these vendors offer circuit review and PCB layout review covering power integrity and signal integrity. So, if it unavoidable to use a canned design, make sure you know the in and out of the design before proceeding with the product implementation.

Points to keep in mind

While the list of points that need to be looked into is big, here are a few key aspects that need attention when the canned design is used:

1. Capture the product requirements completely and map to the canned design you have to ensure there are no gaps. If there are gaps, have a plan to fill them.

2. Spend time to understand the provided reference design so that implementation changes are well understood.

3. Compute the bill of material cost of the reference (canned) design and benchmark it against the budgeted cost. Identify the parts that are expensive and find lower-cost replacement without impacting the performance.

4. If you have to re-lay the PCB to suit your product, understand the thermal and electrical safety and compliance needs and implement. Most reference designs do not need to meet these, so aspects like filtering, heat-sinks, etc have to be properly looked into.

5. If your product is for medical, avionics, or automotive use, remember to generate regulatory documentation for the design and manufacturing for product's approval.

6. Finally, verify and validate the product exhaustively to ensure it has no performance issues. **EFY**



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